EV Technology Course — Module 13: EV Manufacturing and Cost Engineering

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An EV is built in the same four classic shops as a petrol car (press, body, paint, assembly), but the bill of materials (BOM) is very different. The engine, gearbox, fuel and exhaust systems disappear; the battery, e-drive and power electronics appear, and the battery alone can be a third of the vehicle cost. This module applies should-cost and zero-based costing to EVs: where the money goes and how to bring it down.

1. Learning objectives

- Describe how an EV plant differs from an ICE plant.
- Compare EV and ICE cost structures by system.
- Build a should-cost for an e-drive (material, conversion, overhead, profit).
- Apply learning curves, volume and localisation to EV cost reduction.
- Identify VA/VE levers for EV components.
- Understand cost of quality, scrap and capital amortisation in EV programmes.

2. Fundamentals

- **Shops:** press shop (panels), body shop (welding, riveting, bonding, increasingly large castings), paint shop (PT-ED, sealer, primer, topcoat), assembly (trim, chassis, final) plus EV-specific: battery pack line, e-drive line, HV testing.
- **BOM shift:** roughly 30–40% of an EV's cost is the battery pack; e-drive and power electronics add another 10–15%.
- **Fewer moving parts:** an e-drive has a few hundred parts versus well over a thousand in an engine and gearbox, but high-value materials (copper, magnets, semiconductors, cells).
- **Should-cost** = what a part ought to cost given efficient material use, process, cycle time, rates and a fair margin.

3. Detailed technical explanation

What changes in the plant

Area	ICE car	EV
Powertrain	Engine and transmission machining and assembly	E-drive assembly (stator winding, rotor magnet insertion, gearbox), inverter SMT and module assembly
Energy storage	Fuel tank (blow-moulded)	Battery pack: cell receiving, module or cell-to-pack assembly, laser welding, BMS, leak test, EOL
Body	Steel BIW	Battery protection structures, aluminium extrusions, mega/giga castings in some OEMs
Paint	Same process	Same; heavier body may change conveyors and ovens
Assembly	Engine marriage, fluids, exhaust	Battery marriage (bolting pack from below), HV connection, coolant fill, HV safety zones and trained staff
EOL	Emissions, engine run	HV isolation, HVIL, charging test, software flash, dyno run

Cost structure

- Battery cost is driven by cell chemistry (LFP cheaper than NMC), cell price per kWh, and pack integration efficiency (cell-to-pack reduces modules).
- E-drive cost is driven by copper, magnets, electrical steel, aluminium housing, gears and power semiconductors.
- Power electronics cost is dominated by semiconductors (IGBT or SiC) and capacitors.
- Body and interior are similar to ICE, but EVs often have premium features (screens, connectivity) that add cost.

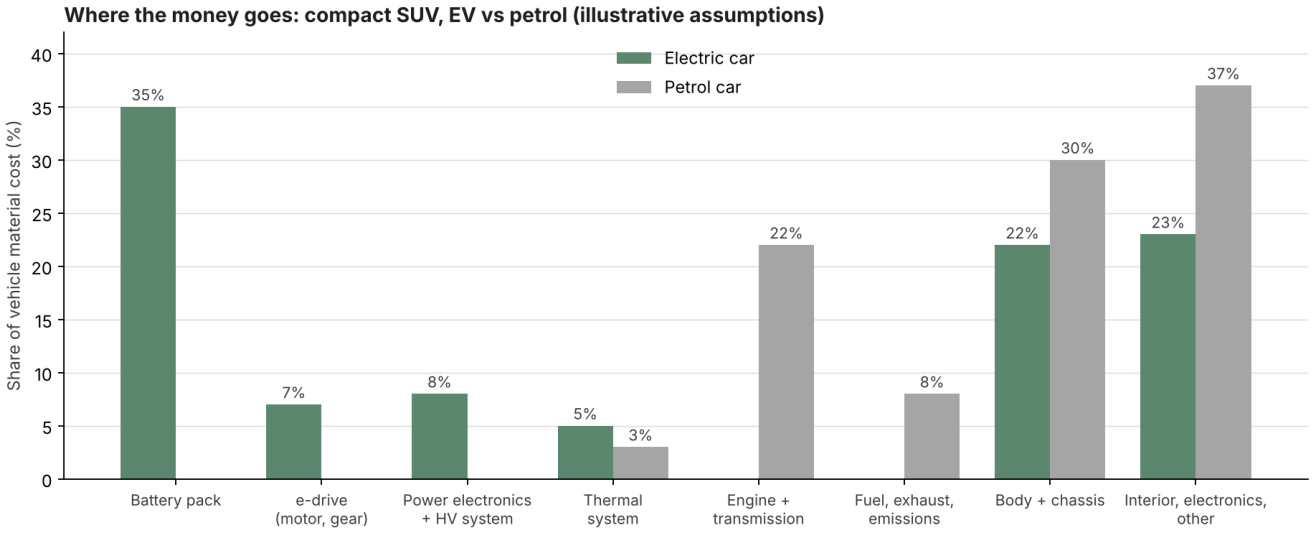
Should-cost method (zero-based)

1. Raw material: gross weight × material rate (include scrap and runner/offcut).
2. Bought-out parts: quoted or benchmarked prices.
3. Conversion: machine hour rate × cycle time ÷ cavities/parts per cycle, plus labour.
4. Overheads: rejection, packaging, logistics, ICC (inventory carrying cost), SG&A.
5. Profit: on value add or on total.
6. Tooling and capex: amortised per unit or paid separately.

Cost-reduction levers

- Chemistry change (NMC → LFP / LMFP), cell-to-pack, larger cells.
- Magnet reduction (reduced-heavy-rare-earth grades, EESM or induction motors), hairpin winding automation.
- SiC versus IGBT trade-off by vehicle class.
- Integration (3-in-1 e-drives, combined OBC/DC-DC, zonal E/E).
- Localisation (India PLI schemes, PMP under PM E-DRIVE), dual sourcing, design-to-cost targets.

4. Diagram: EV vs petrol car cost structure (illustrative)



5. Step-by-step: should-costing a 3-in-1 e-drive

1. Tear down or study the drawing: list motor, gearbox, inverter and housing parts.
2. Weigh or estimate material for each: copper, magnets, laminations, aluminium, steel gears.
3. Apply current commodity rates (copper, NdFeB, CRGO/CRNO steel, aluminium alloy).
4. Add bought-outs: power modules, gate driver PCB, capacitor, sensors, bearings, seals.
5. Estimate conversion: winding, stacking, magnet insertion, machining, gear cutting and grinding, assembly, EOL test cycle times × machine hour rates.
6. Add overheads, rejection and profit.
7. Compare with supplier quote; build a negotiation sheet by cost element.

6. Engineering equations

Part cost:

$$C = C_{material} + C_{BOP} + C_{conversion} + C_{overhead} + C_{profit}$$

Conversion cost:

$$C_{conv} = MHR \times \frac{t_{cycle}}{n_{parts\,per\,cycle}}$$

Learning curve (b = $\log_2$ learning rate):

$$C_V = C_1 \cdot V^{\,b}, \qquad b = \log_2(LR)$$

Capex amortisation per unit:

$$C_{capex} = \frac{\text{Investment}}{\text{years} \times \text{annual volume}}$$

7. Worked examples

Example 1: e-drive should-cost (illustrative, 100 kW class, ₹).

Element	Basis	Cost (₹)
Copper winding	6 kg × ₹900/kg	5,400
NdFeB magnets	1.5 kg × ₹6,000/kg	9,000
Electrical steel (gross)	28 kg × ₹180/kg	5,040
Aluminium housing (HPDC)	12 kg × ₹320/kg	3,840
Gear set, shafts, bearings	Bought-out benchmark	9,000
Power module (IGBT/SiC)	Bought-out benchmark	18,000
Gate driver and control PCB	Bought-out benchmark	6,000
DC-link capacitor, busbars	Bought-out benchmark	4,000
Resolver, sensors, seals, fasteners	Bought-out benchmark	3,500
Material total		**63,780**
Conversion (18% of material)	Winding, machining, assembly, EOL	11,480
Overhead (10%)	Rejection, logistics, SG&A	7,526
Profit (8%)		6,623
Should-cost		**≈ 89,400**

The power module alone is ~20% of the cost, and magnets ~10%: these are the first places to look for savings.

Example 2: conversion cost. Rotor shaft machining with a 4-minute cycle on a ₹2,400/hour machine: 2,400 × 4 ÷ 60 = **₹160** per part.

Example 3: battery share. A 40 kWh pack at ₹9,000/kWh = ₹3.6 lakh, which is **30%** of a ₹12 lakh ex-factory vehicle cost.

Example 4: learning curve. With an 85% learning rate, cost falls to 85% at double cumulative volume, **72%** at 4× and **61%** at 8×.

Example 5: capex amortisation. ₹1,000 crore of tooling and facilities over 5 years at 50,000 vehicles/year = **₹40,000 per vehicle**. Volume matters enormously in EV economics.

8. Real-world examples

- **Tesla** reduced body parts with giga castings and is moving to the "unboxed" assembly approach (building sub-assemblies in parallel).
- **BYD** is vertically integrated (cells, semiconductors, motors), a big reason for its cost advantage.
- **Tata Motors** and **Mahindra** use dedicated EV platforms (acti.ev, INGLO) and local supply chains; Tata sources cells and packs via Tata AutoComp and Agratas plans.
- **India PLI for auto and ACC batteries** and **PM E-DRIVE** reward local value addition; costing engineers must track domestic value addition (DVA) percentages.

9. Manufacturing applications

- **Battery line:** cell grading, stacking, laser welding of busbars, BMS installation, potting, leak test, EOL charge-discharge.
- **E-drive line:** hairpin forming and welding, varnish impregnation, rotor magnet insertion and magnetisation, gearbox assembly, EOL spin and NVH test.
- **HV safety:** trained and certified HV technicians, insulated tools, lockout procedures, isolated test areas.
- **Quality:** weld inspection (vision/X-ray), insulation tests, traceability by serial number for cells and modules.
- **Logistics:** cells are dangerous goods; storage, fire protection and transport rules add cost.

10. Common mistakes and misconceptions

- **"EVs are simpler, so cheaper to build."** Fewer parts, but expensive materials and new capex; cost parity with ICE depends heavily on battery price.
- **"Should-cost is just material plus margin."** Conversion, scrap, logistics, ICC and tooling must all be included.
- **"Commodity prices are fixed."** Copper, lithium and rare earths are volatile; contracts need indexation clauses.
- **"Localisation always saves money."** Only at sufficient volume and with mature suppliers.

11. Practical case study: e-drive cost-down

Problem. An OEM's e-drive quote is ₹1,10,000 against a target of ₹90,000.

Analysis. Should-cost shows ₹89,400. Gaps: the supplier assumes 12% conversion scrap (should be 4%), imported SiC module where IGBT suffices for the vehicle class, and overweight housing.

Actions. Switch to IGBT for the base variant (−₹7,000), reduce housing mass by 1.5 kg through topology optimisation (−₹480), magnet grade change with less dysprosium (−₹1,500), negotiate scrap and conversion (−₹8,000), indexation for copper and magnets.

Lesson. A bottom-up should-cost turns a negotiation into an engineering discussion with numbers.

12. Revision notes

- Battery = 30–40% of EV cost; e-drive + power electronics 10–15%.
- Should-cost = material + BOP + conversion + overhead + profit.
- Conversion = MHR × cycle time ÷ parts per cycle.
- Learning curves and volume drive EV cost down.
- Levers: chemistry, integration, magnet reduction, semiconductors, localisation.

13. Glossary

Term	Meaning
BOM	Bill of materials
BOP	Bought-out parts

Term	Meaning
MHR	Machine hour rate
HPDC	High-pressure die casting
DVA	Domestic value addition
ICC	Inventory carrying cost
VA/VE	Value analysis / value engineering
Learning rate	Cost fraction remaining after each doubling of cumulative volume

14. Knowledge check

Conceptual

1. What new shops or lines does an EV plant need?
2. Why is the battery the biggest cost lever?
3. Why include indexation clauses for EV materials?
4. Why might localisation increase cost at low volume?
5. Name three e-drive VA/VE ideas.

Numerical

1. Machine hour rate ₹1,800, cycle 6 minutes, 1 part per cycle. Conversion cost?
2. You need 10,000 good parts and scrap is 5%. How many must be produced?
3. ₹500 crore capex over 5 years at 40,000 units per year. Amortisation per unit?

Application

1. Build a should-cost structure (headings only) for a battery pack.
2. Propose a cost-reduction roadmap for an EV whose battery is 38% of cost.

15. Answer key

Conceptual

1. Battery pack assembly, e-drive assembly, power-electronics assembly (or supplier), HV test and EOL, battery marriage in assembly, HV safety zones.
2. It is the largest single cost and directly scales with range targets and cell prices.
3. Copper, lithium, nickel and rare-earth prices swing widely; indexation shares risk fairly.

4. Low volumes cannot amortise tooling and supplier overheads, and suppliers may be less mature.
5. Hairpin winding automation, reduced heavy rare earths, IGBT for base variants, integrated housings, gear ratio optimisation, cast-in cooling.

Numerical

1. 1,800 × 6 ÷ 60 = **₹180**.
2. 10,000 ÷ 0.95 = **10,527 parts** (round up).
3. ₹5,000,000,000 ÷ (5 × 40,000) = **₹25,000 per unit**.

Application

1. Cells (count × ₹/cell), modules or CTP structure, busbars and HV connectors, BMS and wiring, thermal system (cold plates, TIM), enclosure and lid, fire and venting parts, fasteners and sealants; conversion (stacking, welding, assembly, EOL formation/test); overheads (scrap, logistics, dangerous-goods handling, warranty provision); profit; tooling.
2. Switch to LFP or LMFP where range permits; cell-to-pack; right-size battery with better efficiency (aero, tyres, heat pump); localise cells and pack parts under PLI; multi-source cells; design for recycling credit; use learning-curve savings with volume growth.

16. Further learning

- BloombergNEF battery price surveys (annual).
- Munro & Associates teardown reports and videos.
- McKinsey and Roland Berger EV cost studies.
- Indian PLI and PM E-DRIVE scheme guidelines (Ministry of Heavy Industries).

Next: Module 14, electric two- and three-wheelers.